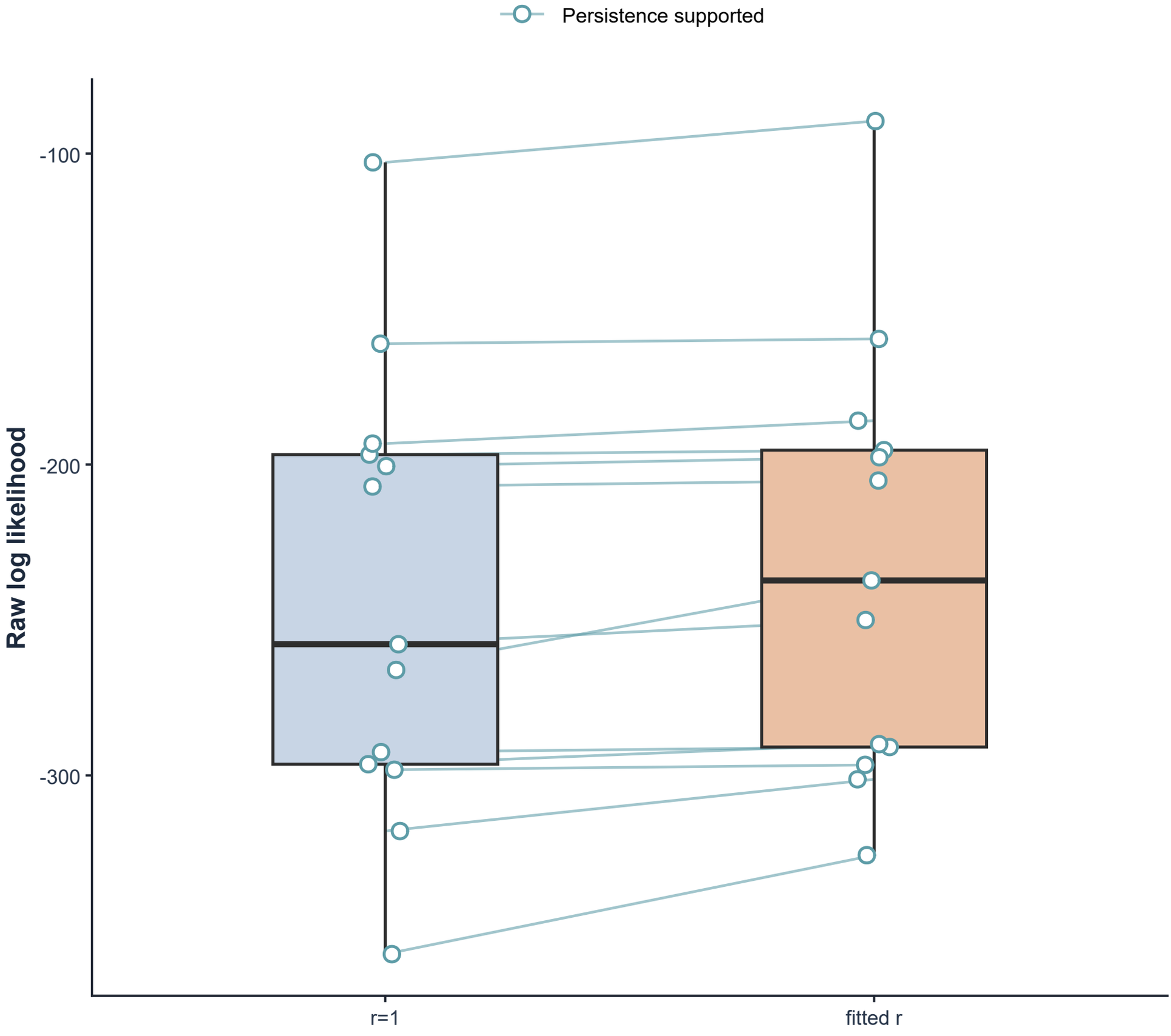logL: r=1 versus fitted-r model